# Swadhin Agrawal

PhD Scholar @Dept. EECS, IISER'B

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## Education

**IISER Bhopal** **Aug. 2020 – Present**

*PhD in Robotics,* *8.67 CPI* *Bhopal, Madhya Pradesh*

*Supervisor: Dr Sujit P.B,*

*Co-supervisor: Dr Andreagiovanni Reina*

**IISER Bhopal** **Aug. 2015 – May 2020**

*BS-MS,* *8.54 CPI* *Bhopal, Madhya Pradesh*

*Physics (major), EECS (minor),*

*Supervisor: Dr Deepak Kumar*

**Gangadhar Meher Junior College** **May. 2013 – May 2015**

*Higher secondary school* *84.5 %* *Sambalpur, Odisha*

## Relevant Coursework

- Control Systems
- Theory of Computation
- Intelligent robotics
- Model verification
- Introduction to ML
- Network Science
- Computational Geometry
- Linear Algebra
- Math Methods I II
- Multivariable Calculus
- Probability and Statistics
- Behavioral Biology
- Self-Driving Cars
- SLAM

## Technical Skills

**Languages**: Python, C, C++, C, Matlab

**Developer Tools**: VS Code, Blender, OpenFoam

**Technologies/Frameworks**: Linux, GitHub, ROS 1, ROS 2, Docker Containers

**Simulators**: Unity, Gazebo, Webots

**Python libraries**: Statmodels, Pandas, Numpy, Matplotlib, Multiprocessing, Numba, Ray

## Teaching Activities

**IISER Bhopal** **Aug 2022 – Present**

*Teaching Assistant,* *Bhopal, MP*

*Computer Organization ECS409*

**IISER Bhopal** **Jan 2021 – May 2022**

*Teaching Assistant,* *Bhopal, MP*

*Data structures and algorithms ECS202*

**IISER Bhopal** **Aug 2021 – Nov 2021**

*Teaching Assistant,* *Bhopal, MP*

*Control systems ECS323*

**IISER Bhopal** **Jan 2021 – May 2021**

*Teaching Assistant,* *Bhopal, MP*

*Introduction to programming ECS102*

**IISER Bhopal** **Aug 2020 – Nov 2020**

*Teaching Assistant,* *Bhopal, MP*

*Control systems ECS323*

## Academic Activities

**IISER Bhopal** **May 2019 – May 2020**

*MS Thesis,* *Bhopal, MP*

*Effect of Gaussian curvature on frictional interaction between an ultra-thin film and a soft-substrate*

∗ In this project, we explored the role of geometry in modifying the strength of adhesion and hence friction between a few nano-meter thin polystyrene film and a soft hydrogel substrate using RGB image processing techniques on acquired data through long timed imaging.

- ∗ In our experiments, we loaded the ultra-thin films on top of the swollen hydrogel surfaces that shrink over time due to the homogeneous evaporation of water.
- ∗ By measuring the compression of the film and the shrinking rate of the surface through image analysis using MATLAB, we found that the frictional interaction between the substrate and the film depends on the underlying Gaussian curvature of the substrate.
- ∗ As part of the thesis, I also designed the automated experimental setup using Laser and (Olympus & Nikkon) DSLR camera, both controlled through Arduino UNO.

**Robert Bosch Center for Cyber-Physical Systems** **May 2019 – July 2019**
*Intern under Dr Raghu Krishnapuram,* *Bangalore, Karnataka*
*Depth estimation from Monocular image sequences using machine learning methods*

- This project was a part of the MBZIRC 2020 competition. In this project, our target was to use a monocular camera and estimate the depth of the objects in the scene.
- We explored various kinds of existing methods like using Convolutional Neural Networks, Geometrical approaches and Dense Nets.
- We tried to reinvent Struct2Depth by Google Brain, DEEPTAM and Demon by LMB-Freiburg, SFMLearner by Tinghui Zhou et al., Disparity learning due to time (in SFMLearner) as well as Space Disparity Learning (MonoDepth by Godard et al.).

**IISER Bhopal** **June 2018 – July 2018**
*Intern under Dr Kushal Kumar Shah,* *Bhopal, MP*
*On finding the exact Analytical solution of Fokker Planck Equation for Plasma*

- In this project, we made an attempt to find the exact solution for the Fokker Planck Equation derived for a Paul trap which was a higher-order higher degree multivariable partial differential equation.
- This equation gives rise to Riccati Equations, which we did rethinking and made an attempt to solve numerically in Python in order to understand the possibilities of finding the exact solutions. Results of this project can be found in the website link provided above.
- As a part of the internship, we also presented poster of our work at the Plasma conference held at University of Delhi, in 2018. Our work has also been acknowledge as the preliminary work which was published in AIP Physics of Plasma.

**Poornaprajna Institute of Scientific Research** **May 2017 – July 2017**
*Intern under Dr R Srikant,* *Bangalore, Karnataka*
*On Approximating Universal Quantum Computation*

- Here, I learned the basic aspects of Qubits and Quantum Computation, Quantum Gates and Circuits.
- Also, I used the IBM Quantum Gates and created Quantum gates in Mathematica using a software package called QETLAB.
- I explored Quantum Computation and Quantum Information from McMahon and Nielsen and Chuang's books.

## Challenges

**MBZIRC 2023** | *Semi-Finalist Team LUNA , Student Leader* **Dec 2021 - Aug 2022**

- Implemented the decentralized swarm motion algorithm for inspection and payload lifting from the target vessel in ROS2, Gazebo Ignition MBZIRC environment and Docker.
- Implemented pick and place task for the Robotic arm.
- Implemented lawn-mower algorithm for target vessel search on a delta-fixed-wing vehicle.
- Implemented PID controlled go-to-goal algorithm for unmanned surface vehicle.

**VORC 2020** | *Team Unmet dependencies, Student Leader* **Nov 2020 - Dec 2020**

- Participated in the competition to get exposure to the latest software frameworks like Docker and ROS1.

## Scientific Reviews

Swarm Intelligence | *Special Issue, Springer* **June 2021**

COCOON 2022 | *Conference* **2022**

## Publications

### Conferences

1. Swadhin Agrawal, Sujit P. Baliyarasimhuni, and Andreagiovanni Reina. Response threshold distributions to improve best-of-n decisions in minimalistic robot swarms. In *Thirteenth International Conference on Swarm Intelligence (ANTS 2022)*

## Fellowships and Distinctions

**Junior Research Fellowship** **Aug 2020 – Present**
*Institute* *IISER'B*

**Innovation in Science Pursuit for Inspired Research Fellowship** **Aug 2015 – May 2020**
*DST-Inspire* *IISER'B*

**Student of the Year** **2015**
*Principal* *Gangadhar Meher Junior College*

## Societies & Memberships

**IEEE Robotics & Automation Society (RAS)** **2020 - Present**
*Student Member*

**Society for the Study of Evolution (SSE)** **2020 - Present**
*Student Member*

## Extracurricular

**Junior Research Fellowship** **Aug 2020 – Present**
*Institute* *IISER'B*

**Innovation in Science Pursuit for Inspired Research (DST-INSPIRE) Fellowship** **Aug 2015 – May 2020**
*DST-Inspire* *IISER'B*

**Student of the Year** **2015**
*Principal* *Gangadhar Meher Junior College*

## Leadership

**Secretary** **Nov 2017 - May 2018**
*Sports Council* *IISER'B*

**Coordinator** **2016**
*Computer Science & Programming Club* *IISER'B*

**Founder & Coordinator** **2018-2020**
*Rocket & Satellite Designing Club* *IISER'B*

**Student Leader** **2021-2022**
*MBZIRC 2023* *Team LUNA, IISER'B*

## Awards & FunFacts

**Secretary** **Nov 2017 - May 2018**
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*Rocket & Satellite Designing Club* *IISER'B*

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